

PUBLISHED: 08-JUN-2017
2014.0 RANGE ROVER (LG), 303-04

FUEL CHARGING AND CONTROLS - V8 S/C 5.0L PETROL

RIGHT FUEL RAIL (G1509467)

REMOVAL AND INSTALLATION

19.60.05	FUEL RAIL - RH BANK - RENEW	5000 CC, AJ V8, SUPERCHARGED	3	USED WITHINS	
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SPECIAL TOOL(S)



E114526

310-197

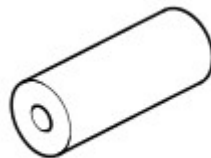
Remover, Fuel
Injector



E107680

310-198

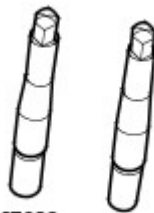
Installer,
Teflon Seal



E107681

310-199

Re-shape Tool, Teflon
Seal



E107682

310-200-01

Fuel Rail Installation Guide
Pins - Threaded

PART(S)

STEP	PART NAME	QUANTITY
Installation Step 1	Fuel rail LH supply pipe	1
Installation Step 1	Fuel rail RH supply pipe	1
Installation Step 8	Fuel injector seal	1
Installation Step 9	Fuel injector seal	1
Installation Step 11	Fuel Injector replacement kits	1

REMOVAL



CAUTIONS:

- Make sure that tools and equipment are clean and free of foreign material and lubricant.
- Always carry out the cleaning process before carrying out any repairs to the fuel injection system components. Failure to follow this instruction may result in foreign matter ingress to the fuel injection system.



NOTES:

- Some variation in the illustrations may occur, but the essential information is always correct.
- Some components shown removed for clarity.
- Removal steps in this procedure may contain installation details.

1.

Refer to: Petrol and Petrol-Ethanol Fuel Systems Health and Safety Precautions (100-00 General Information, Description and Operation).

2.

Refer to: Fuel System Pressure Release - GTDi 2.0L Petrol/V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol (310-00 Fuel System - General Information, General Procedures).

3.

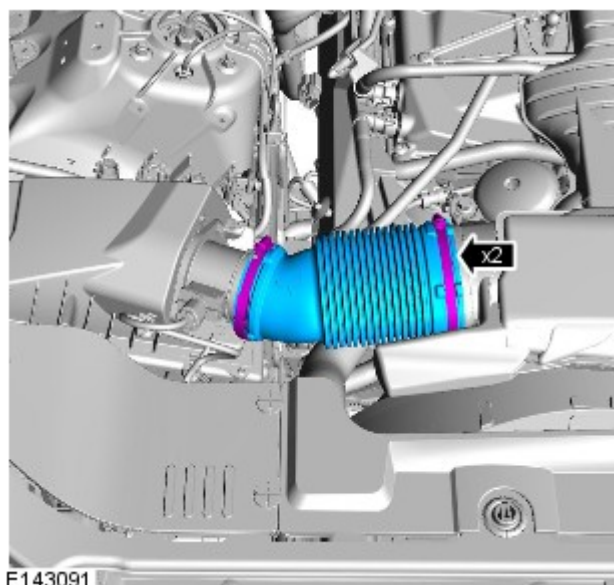
Refer to: Engine Cover - V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol (501-05 Interior Trim and Ornamentation, Removal and Installation).

4. Disconnect the battery ground cable.

Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications).

5. Refer to: Air Cleaner Outlet Pipe T-Connector (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

- 6.



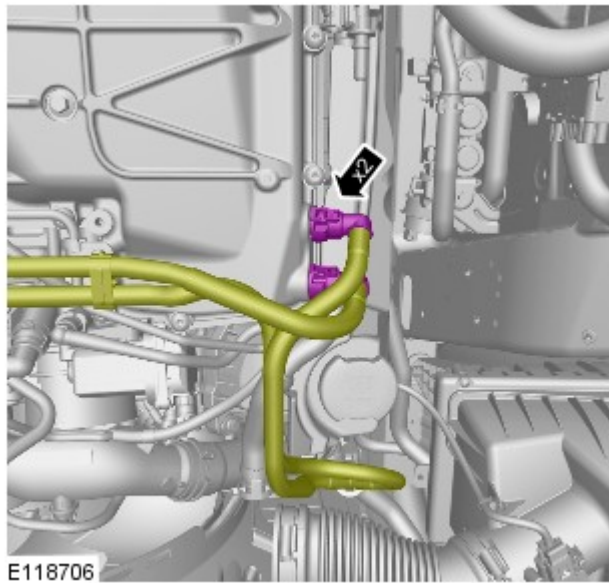
7. Refer to: Right Air Cleaner (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation).

- 8.

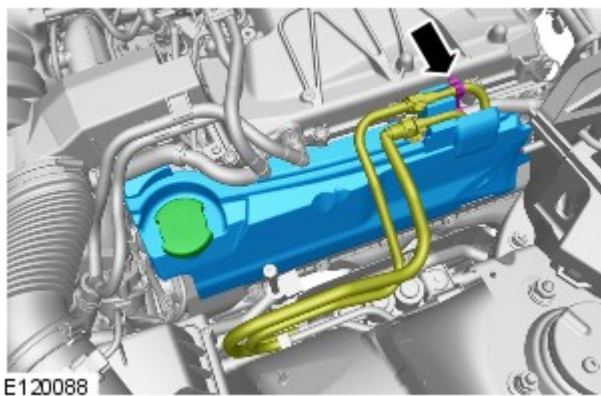


CAUTION:

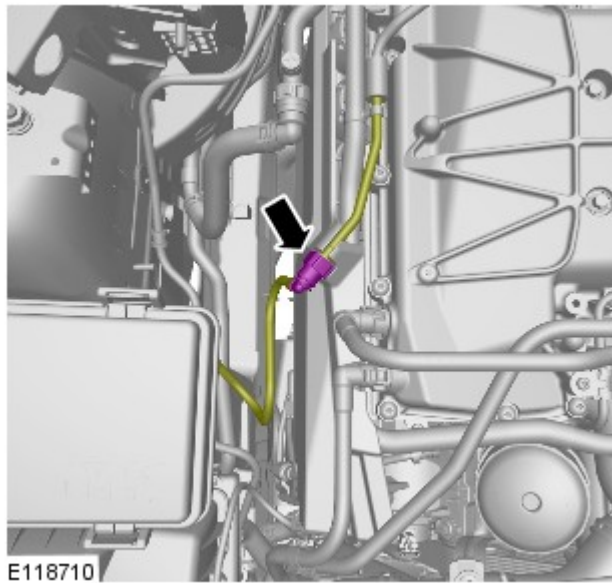
Be prepared to collect escaping coolant.



9.



10.

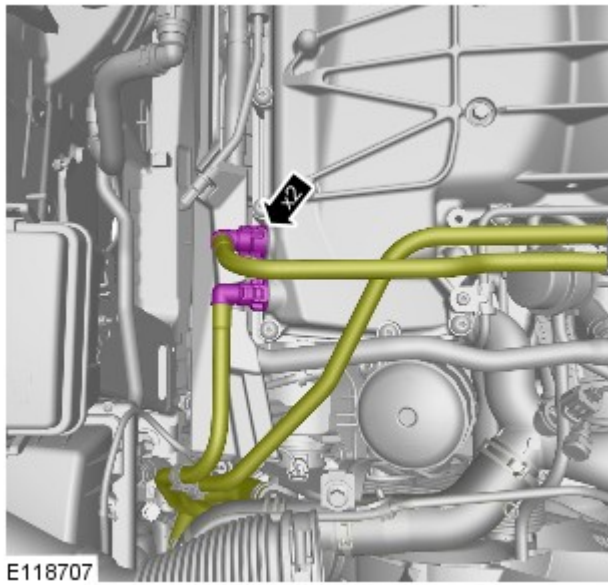


11.

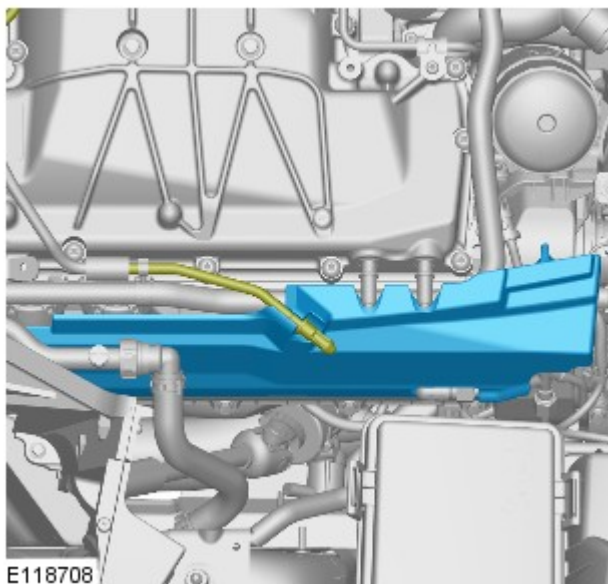


CAUTION:

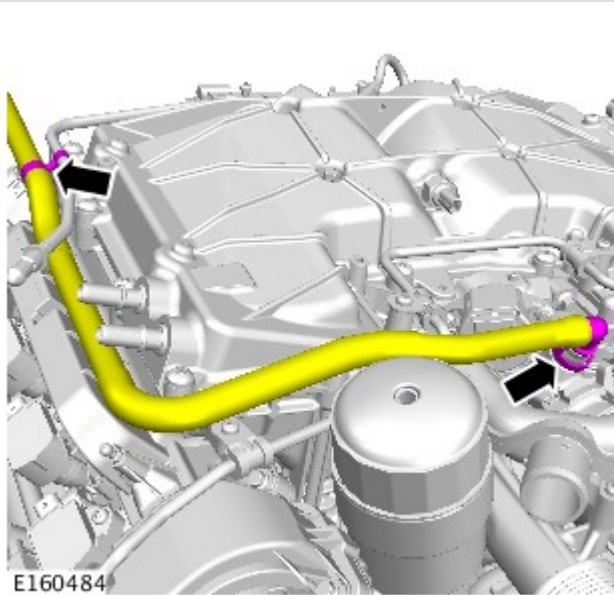
Be prepared to collect escaping coolant.



12.



13.



14.



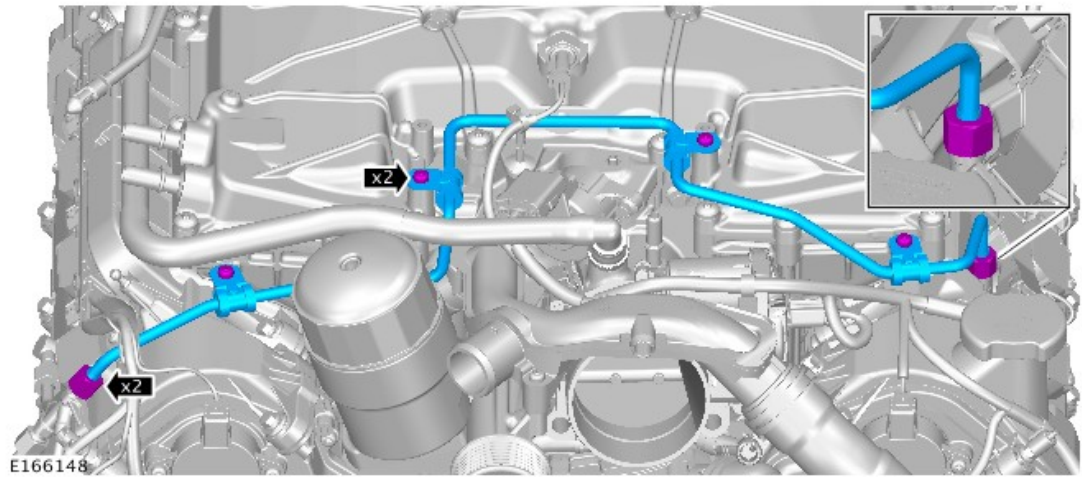
WARNING:

Do not smoke or carry lighted tobacco or open flame of any type when working on or near any fuel related components. Highly flammable vapors are always present and may ignite. Failure to follow these instructions may result in personal injury.

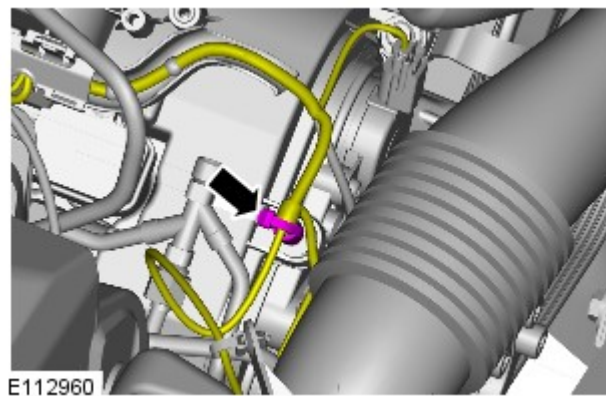


CAUTIONS:

- Make sure that the fuel line union does not rotate.
- Be prepared to collect escaping fuel.
- Make sure that all openings are sealed. Use new blanking caps.
- This fuel pipe can be re-used, inspect for signs of damage. Renew if necessary.



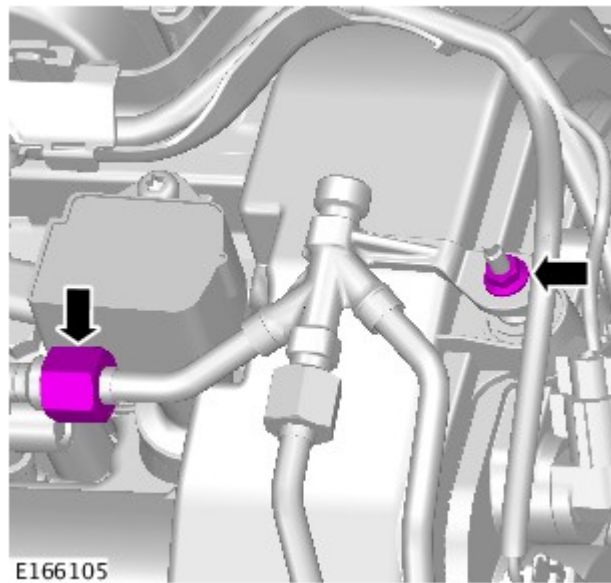
15.



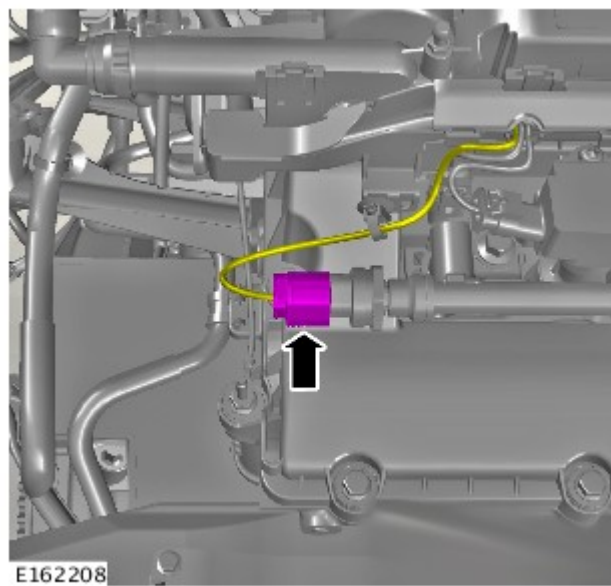
16.

⚠ CAUTIONS:

- Be prepared to collect escaping fuel.
- Make sure that all openings are sealed. Use new blanking caps.



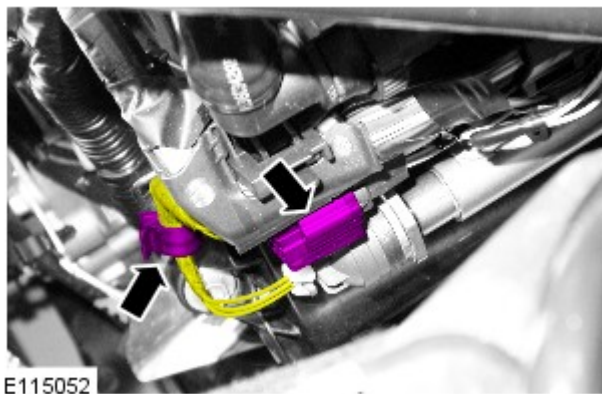
17.



18.

△ NOTE:

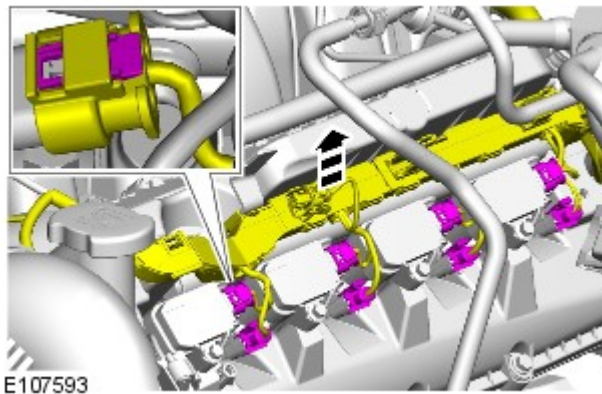
If equipped.



19.

 **NOTE:**

LH illustration shown, RH is similar.



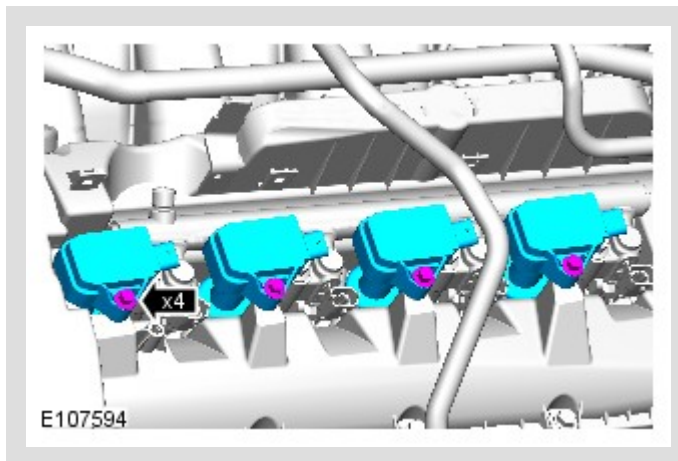
20.

Refer to: Fuel Injection Component Cleaning (303-04 Fuel Charging and Controls - V6 S/C 3.0L Petrol, General Procedures).

21.

 NOTE:

LH illustration shown, RH is similar.



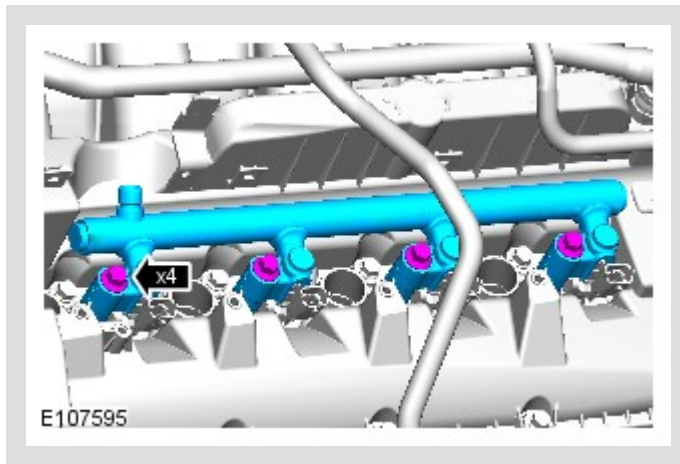
22.

 CAUTION:

Make a note of the fuel injector clamp alignment to the fuel rail prior to removal.

 NOTE:

LH illustration shown, RH is similar.



23.

 NOTE:

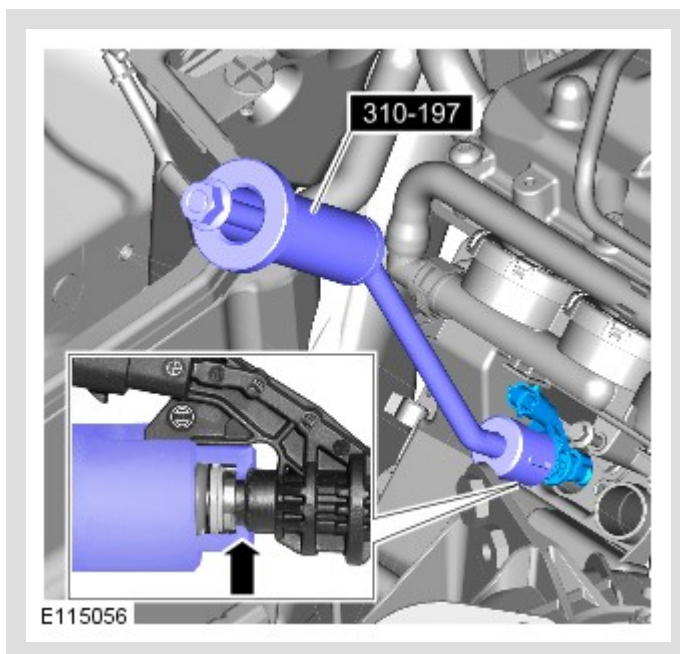
This step is only required if a new fuel rail is installed.

Refer to: Fuel Rail Pressure Sensor (303-14 Electronic Engine Controls - V8 N /A 5.0L Petrol, Removal and Installation).

24.

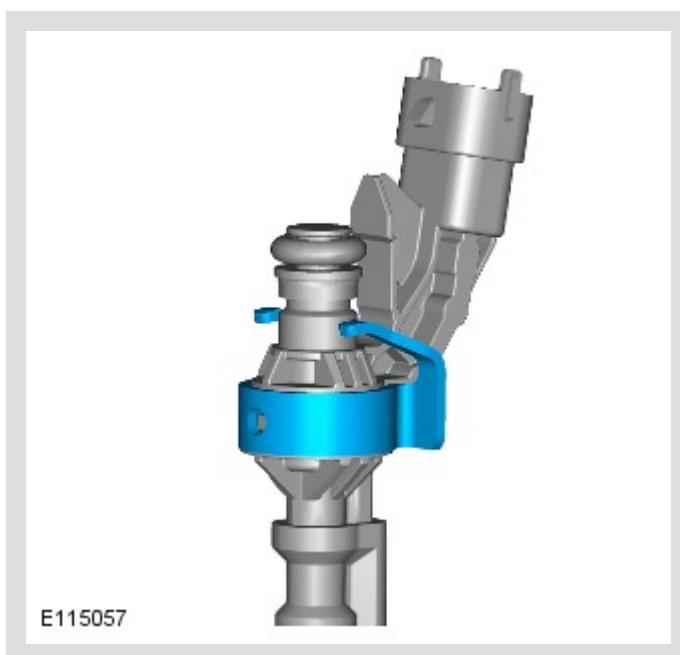
 CAUTIONS:

- Make sure that the special tool is located correctly to the fuel injector prior to removing the fuel injector.
- Make sure that the special tool is held square to the fuel injector during removal.
- Make sure that all open ports are covered to prevent any foreign material ingress.



Special Tool(s): [310-197](#)

25.



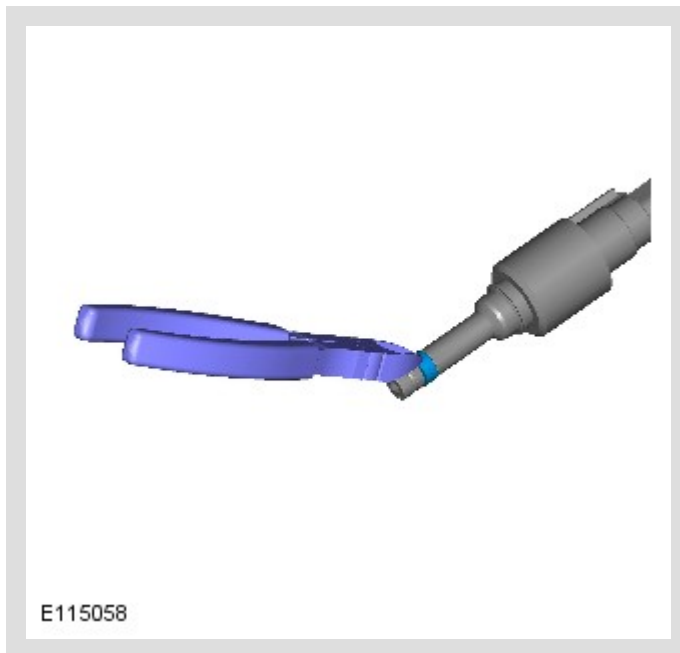
Remove and discard the fuel injector clamp.

26.



CAUTIONS:

- Do not use a knife to remove the Teflon seal as damage could occur to the fuel injector.
- Do not cut the Teflon seal too deep as damage could occur to the fuel injector.
- Pinch the Teflon seal to allow the tool to cut the Teflon seal without damaging the fuel injector.



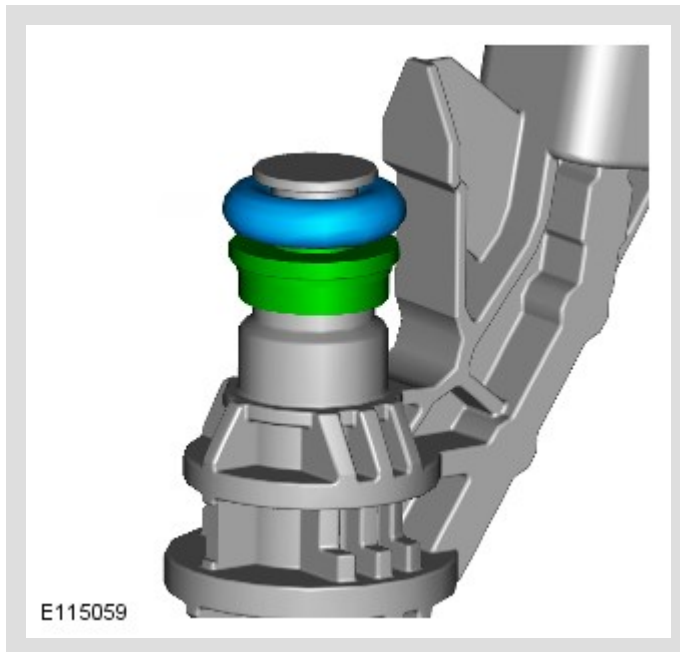
Remove and discard the fuel injector seal.

27.



CAUTION:

Do not use any sharp tools to remove the O-ring seal as damage could occur to the fuel injector.



Remove and discard the fuel injector O-ring seals.

28.



WARNING:

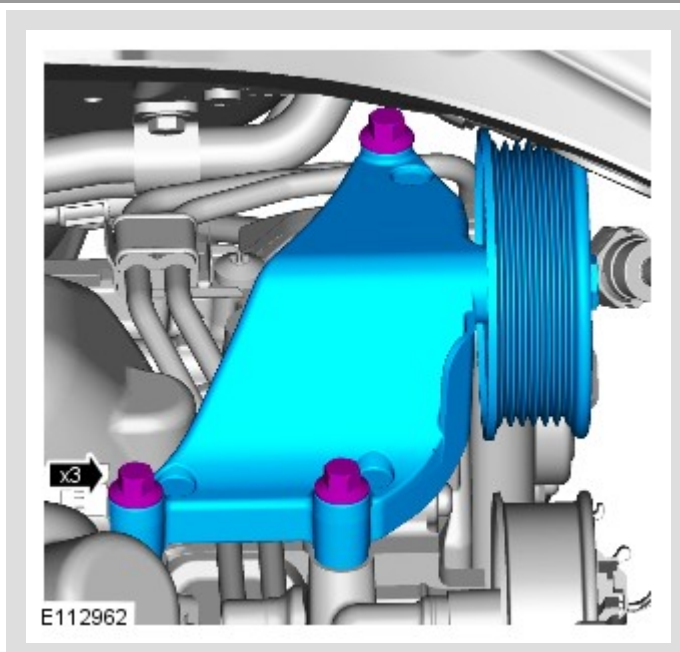
Do not work on or under a vehicle supported only by a jack. Always support the vehicle on safety stands.

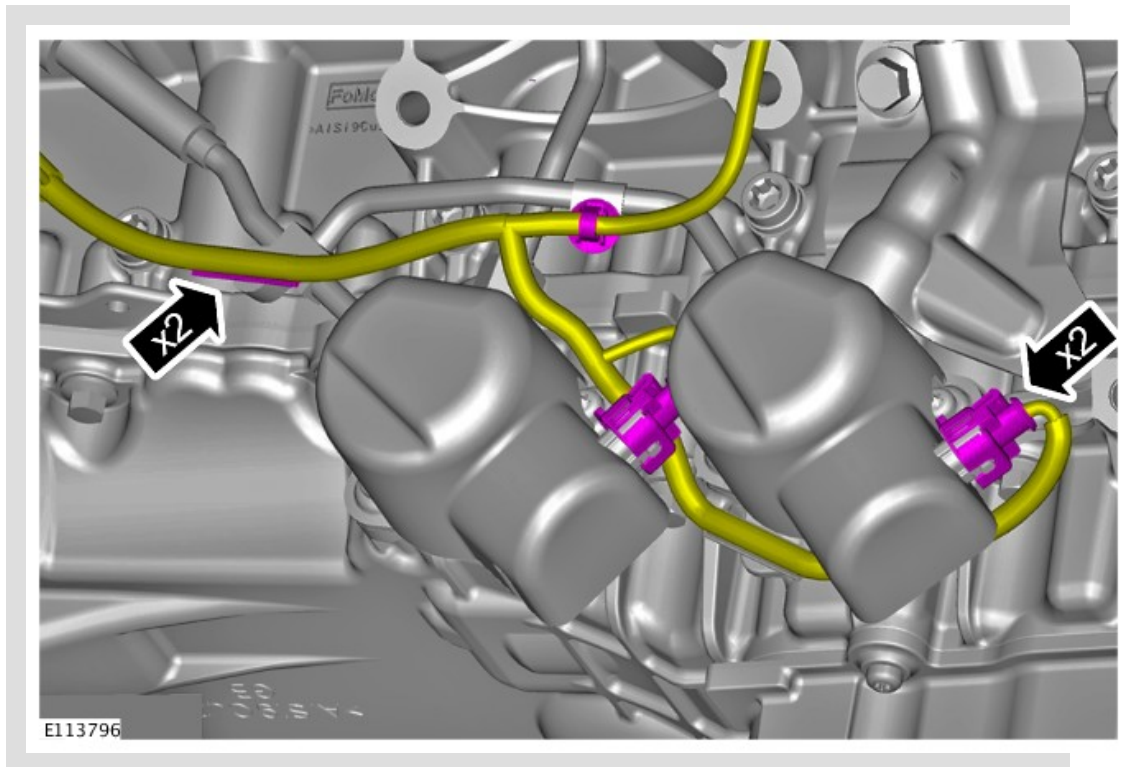
Raise and support the vehicle.

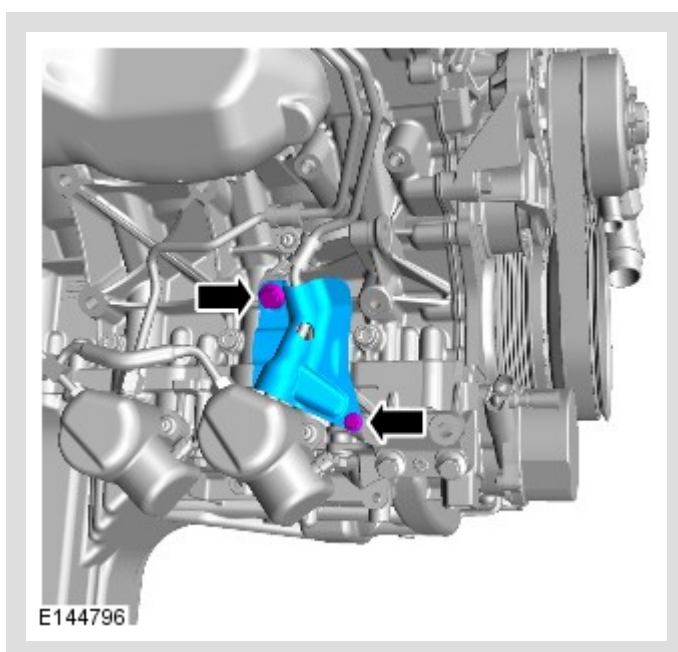
29.

Refer to: Generator (414-02 Generator and Regulator - V6 S/C 3.0L Petrol, Removal and Installation).

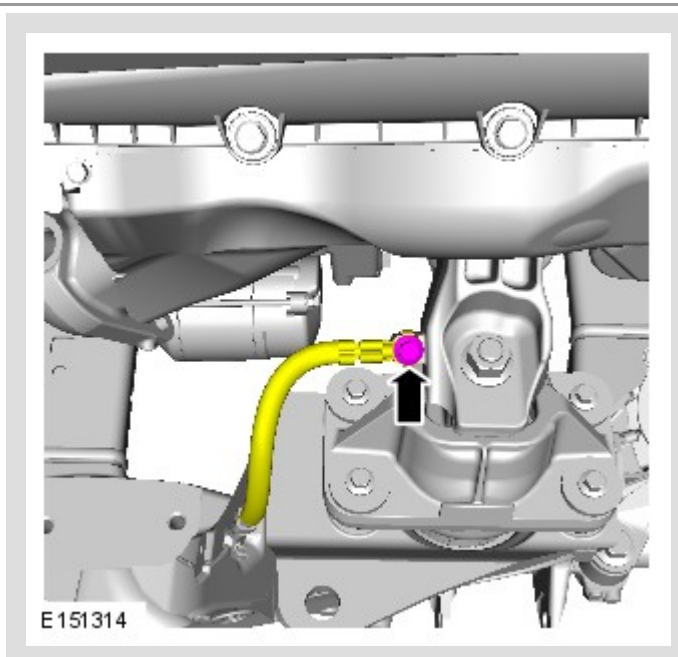
30.



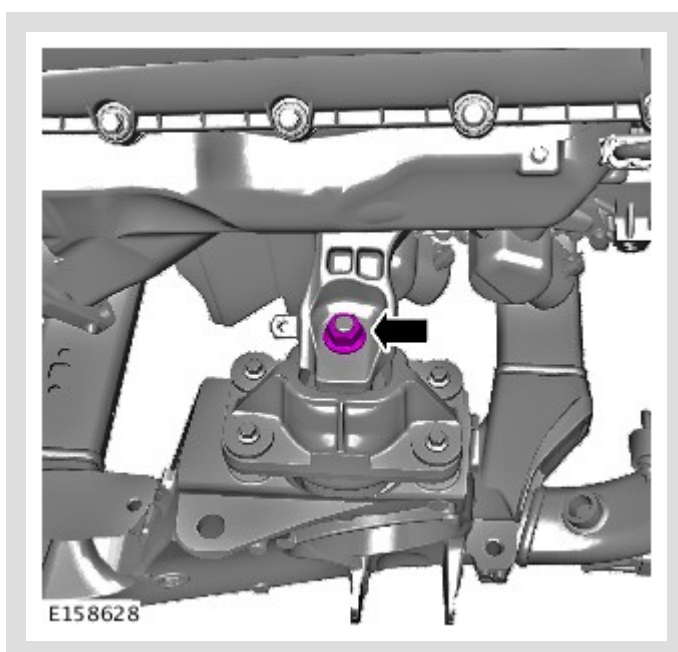




33.



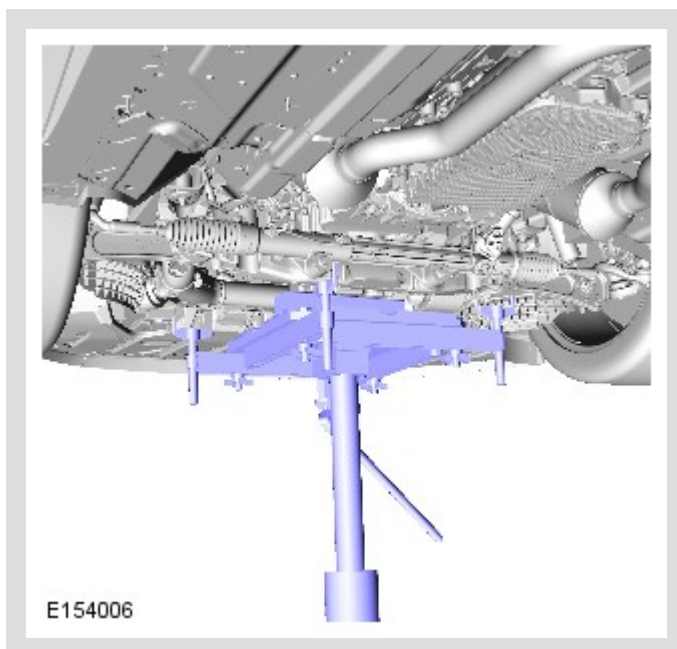
34.



35.

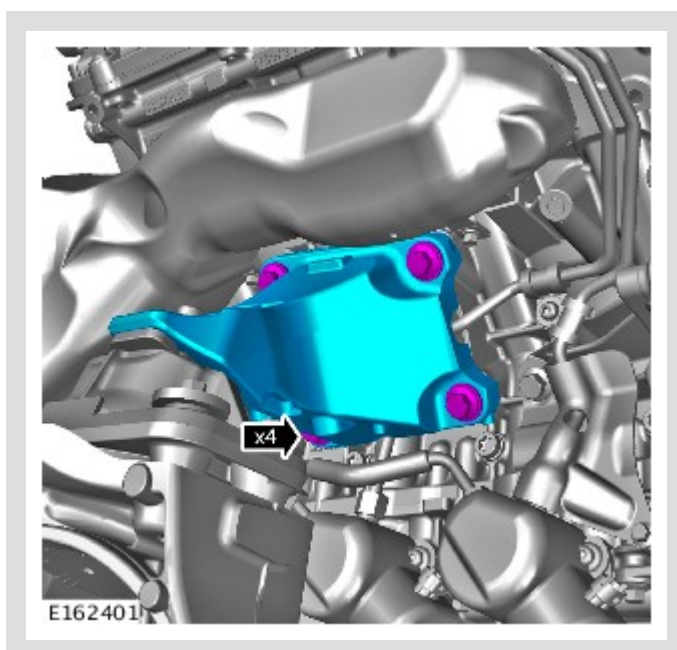
ⓘ CAUTION:

Use a wooden block to protect the oil pan when supporting the engine.



Using a suitable hydraulic jack, support the engine.

36.



37.



WARNING:

Do not smoke or carry lighted tobacco or open flame of any type when working on or near any fuel related components. Highly flammable vapors are always present and may ignite. Failure to follow these instructions may result in personal injury.



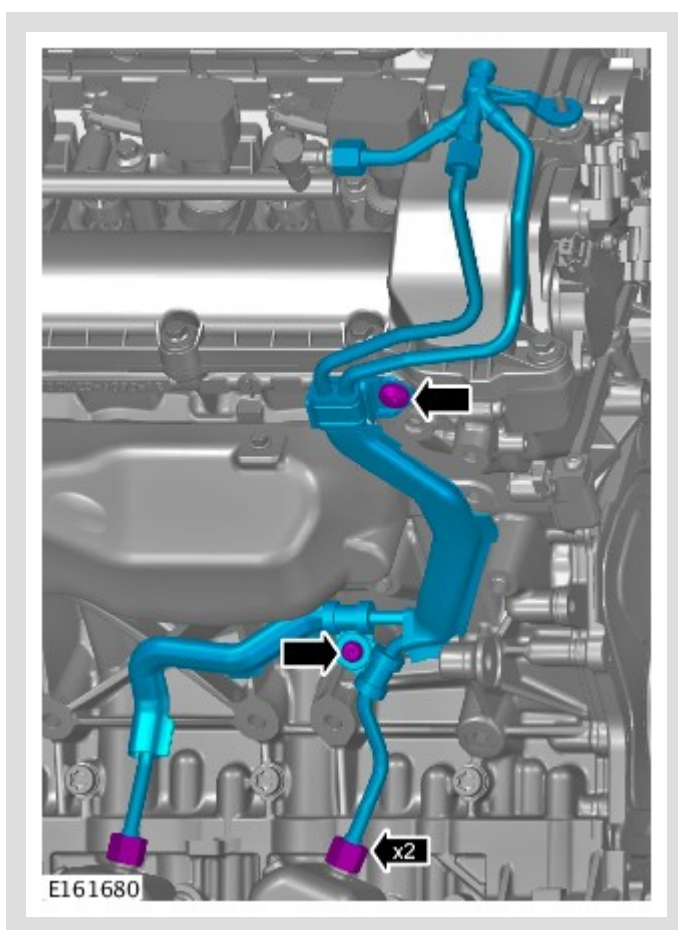
CAUTIONS:

- Remove and discard the high-pressure fuel supply lines.
- Be prepared to collect escaping fluids.
- Make sure that all openings are sealed. Use new blanking caps.



NOTE:

Engine shown removed for clarity.



INSTALLATION

1.

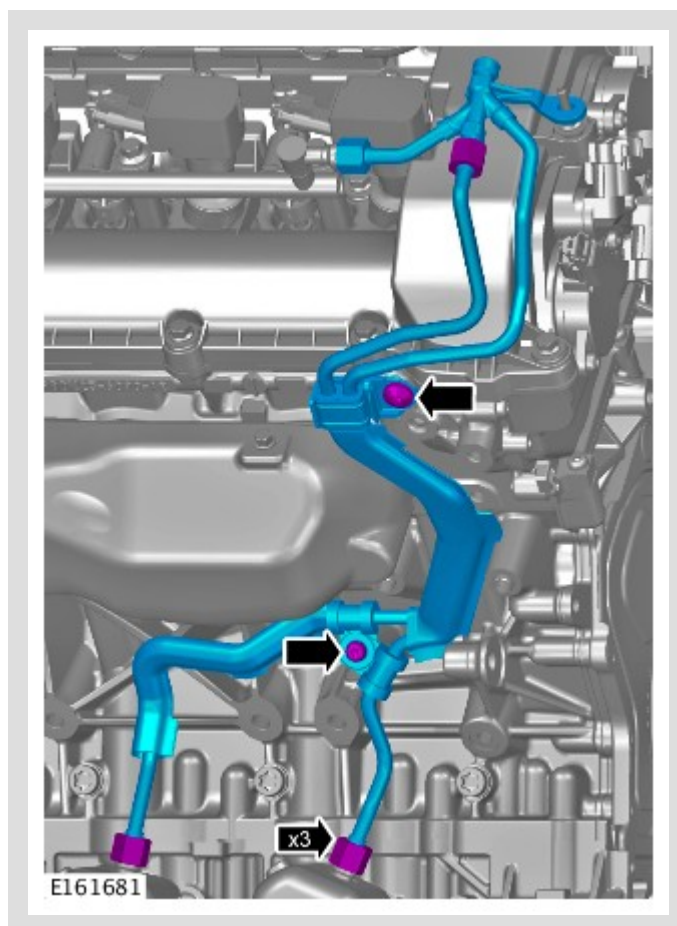
⚠ CAUTION:

Lubricate only the union threads with clean engine oil.

△ NOTES:

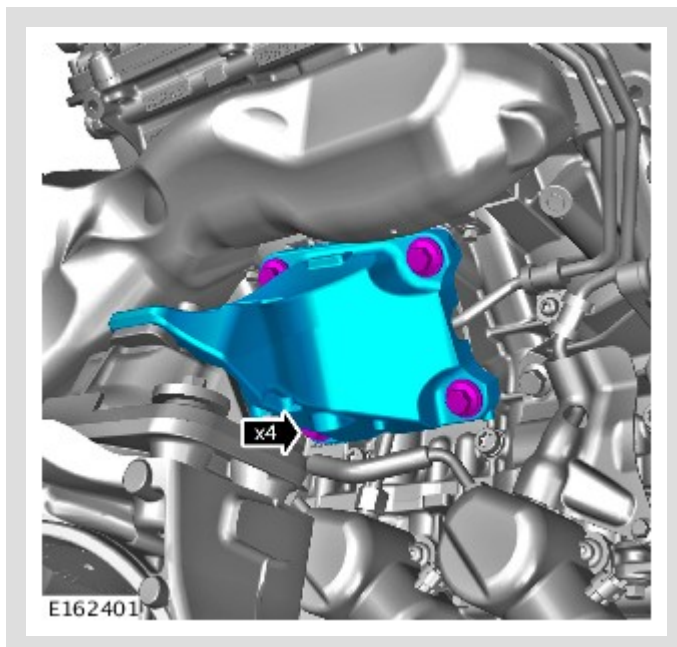
- Engine shown removed for clarity.

- Install the bolts and unions finger tight before final tightening.
- Remove and discard the blanking caps.



Renew Part: *Fuel rail LH supply pipe* Quantity: 1 .

Renew Part: *Fuel rail RH supply pipe* Quantity: 1 .

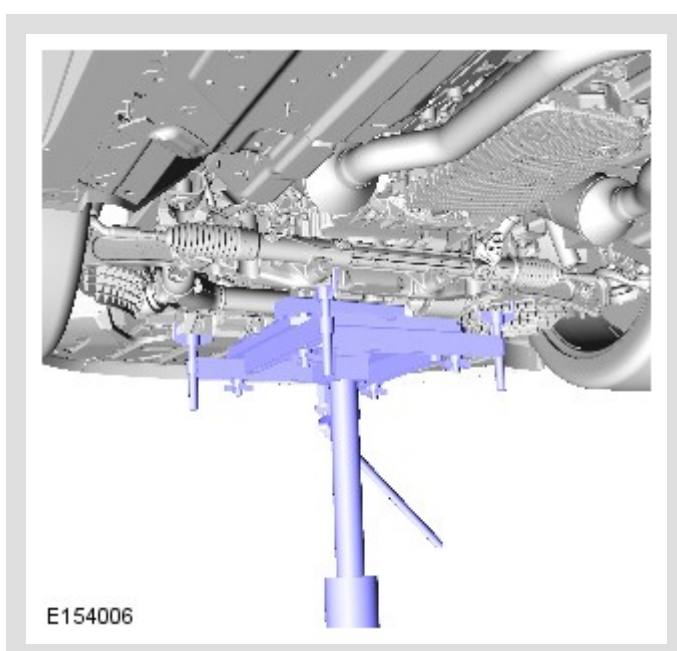


Torque:

Stage 1: **45 Nm**

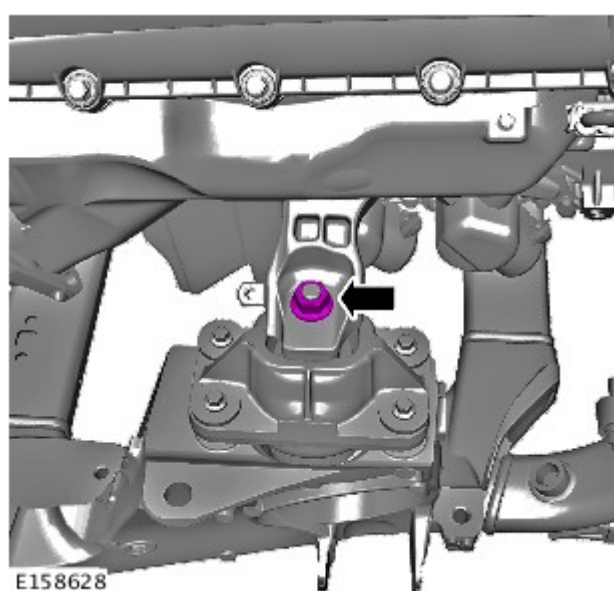
Stage 2: **60°**

3.



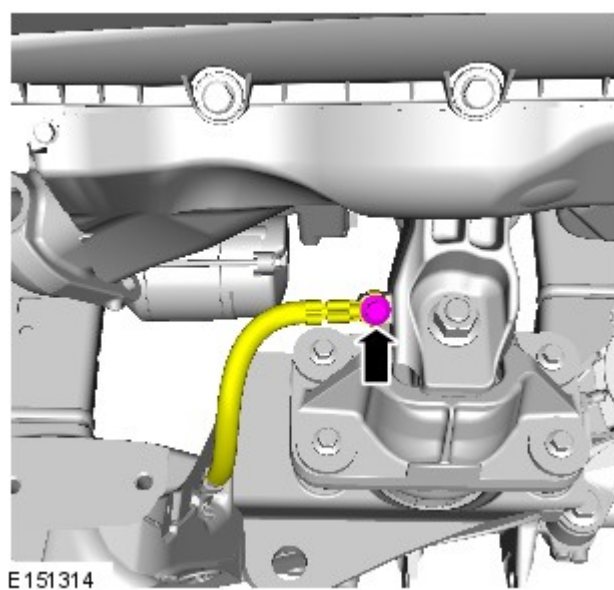
Remove the special tool.

4.



Torque: 130 Nm

5.

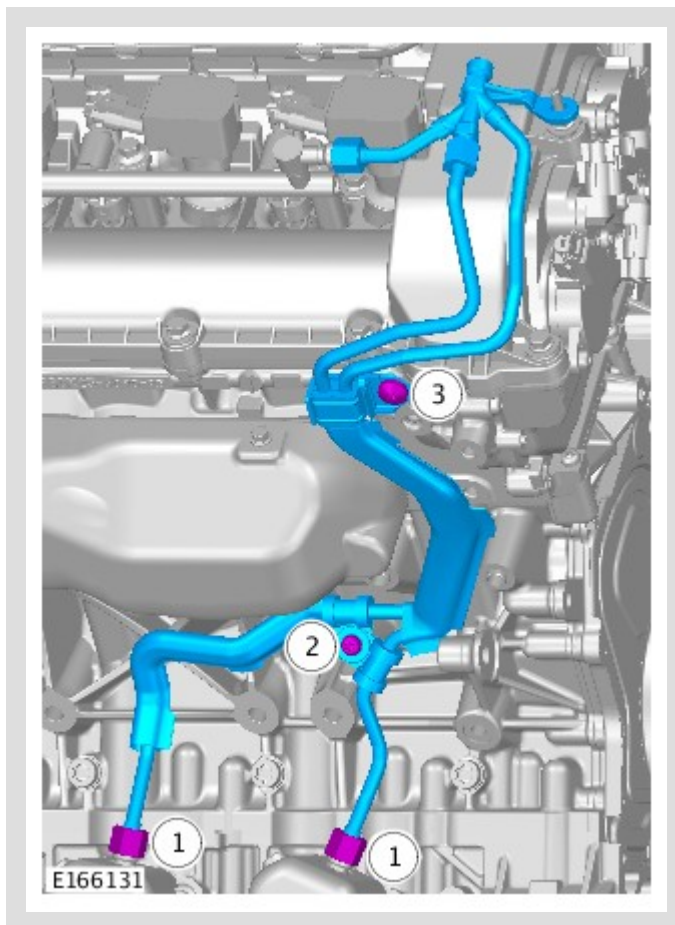


Torque: 10 Nm

6.

 NOTE:

Some components shown removed for clarity.



Torque:

Unions (1) **21 Nm**

M6 (2) **11 Nm**

M8 (3) **25 Nm**

7.

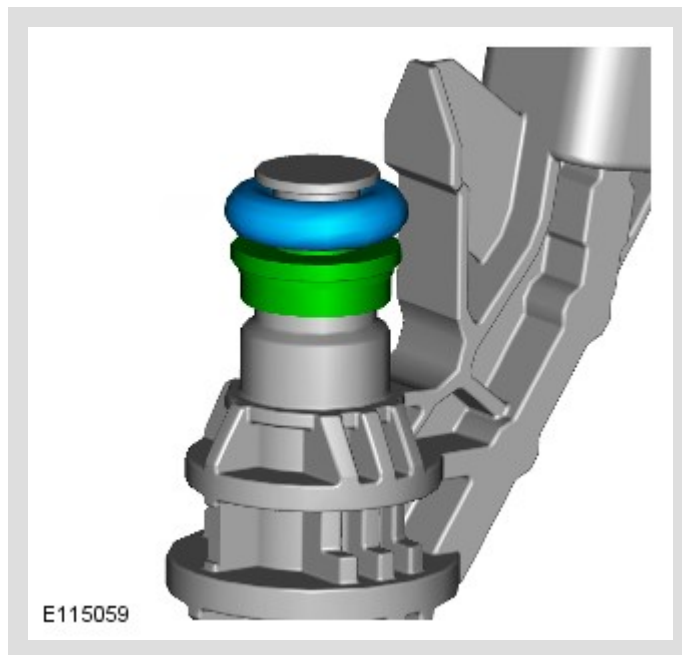
Lower the vehicle.

8.



NOTE:

Repeat this step for the remaining fuel injectors.



Install new fuel injector O-ring seals.

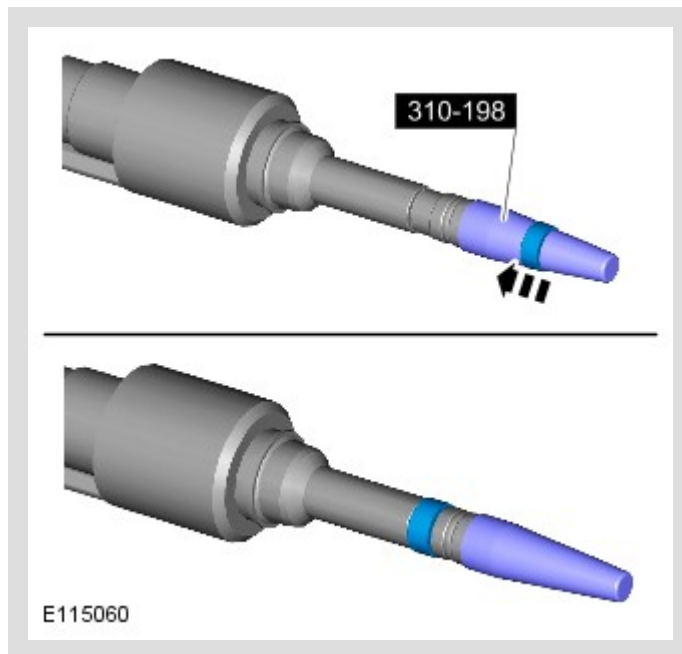
Renew Part: *Fuel injector seal* Quantity: *1* .

9.



NOTE:

Repeat this step for the remaining fuel injectors.

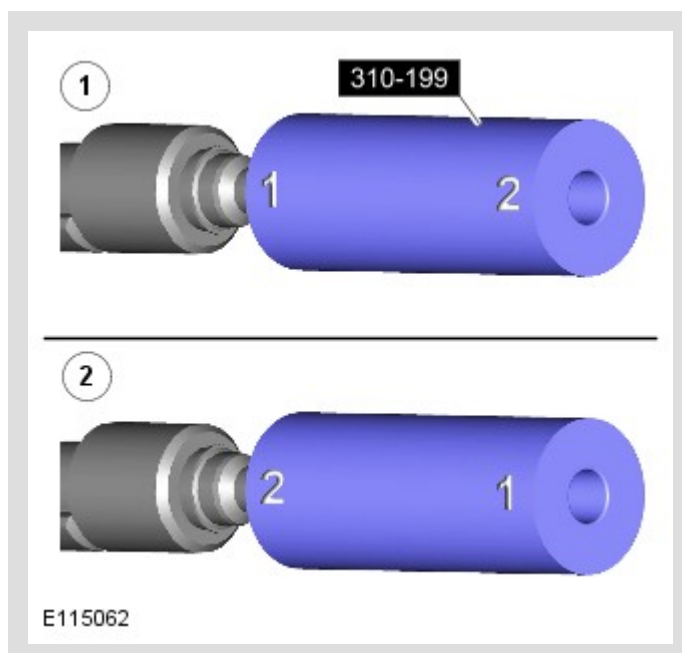


Install new Teflon seals on the lower tips of the injectors.

Special Tool(s): 310-198

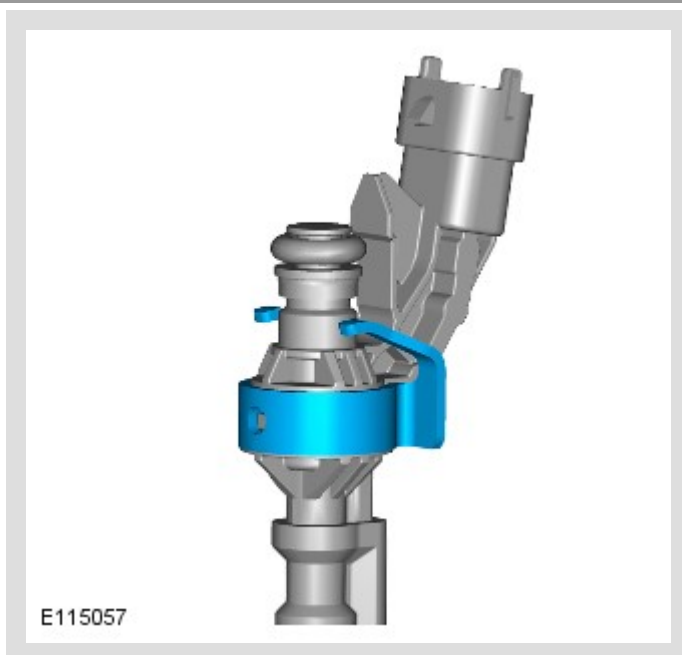
Renew Part: *Fuel injector seal* Quantity: 1 .

10.



Special Tool(s): 310-199

11.



Install a new fuel injector clamp.

Renew Part: *Fuel Injector replacement kits* Quantity: 1 .

12.



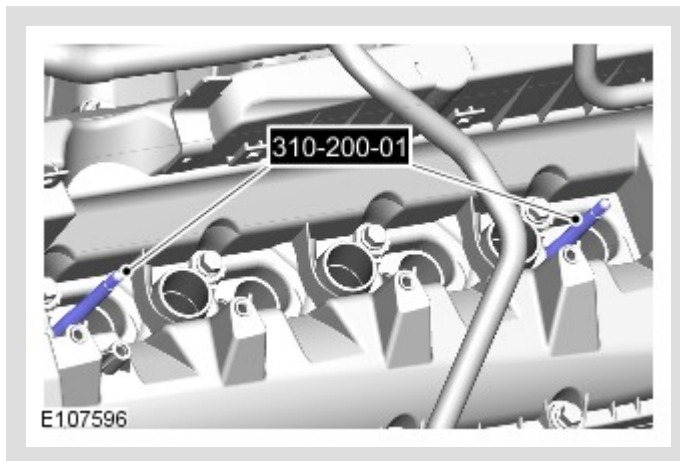
CAUTION:

If a new cylinder head has been installed then the special tool 310-200-02 without the thread must be used to install the fuel rail.



NOTE:

LH illustration shown, RH is similar.



Special Tool(s): [310-200-01](#)

13.



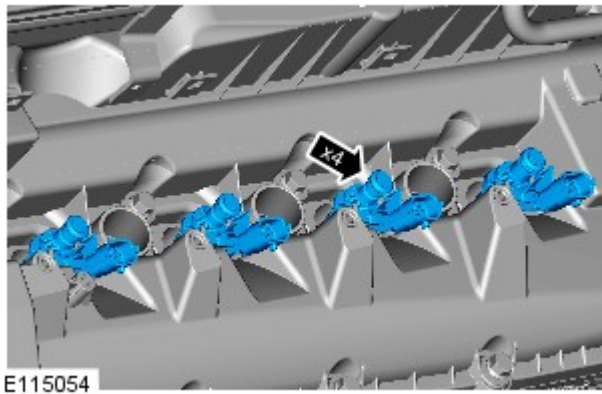
CAUTIONS:

- Make sure that the area around the open fuel injector ports are clean and free of foreign material and lubricant prior to installing the fuel injector.
- When installing the fuel injector(s), make sure that the Teflon seal is clean and free of foreign material and lubricant.



NOTE:

LH illustration shown, RH is similar.



14.

 NOTE:

This step is only required if a new fuel rail is installed.

Refer to: Fuel Rail Pressure Sensor (303-14 Electronic Engine Controls - V8 N/A 5.0L Petrol, Removal and Installation).

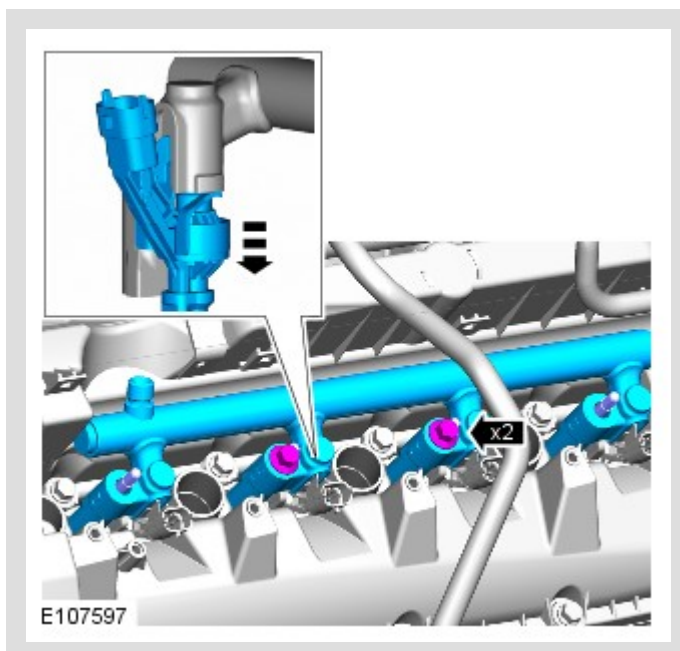
15.

 CAUTIONS:

- If new fuel injectors are installed, a new injector clamp must be installed.
- Make sure that the fuel injector is aligned and installed into the fuel rail correctly, as noted in the removal step.
- Tighten the fuel rail retaining bolts a turn at a time until the correct torque is achieved.

 NOTES:

- Lubricate the fuel injector O-ring seals with clean engine oil.
- LH illustration shown, RH is similar.

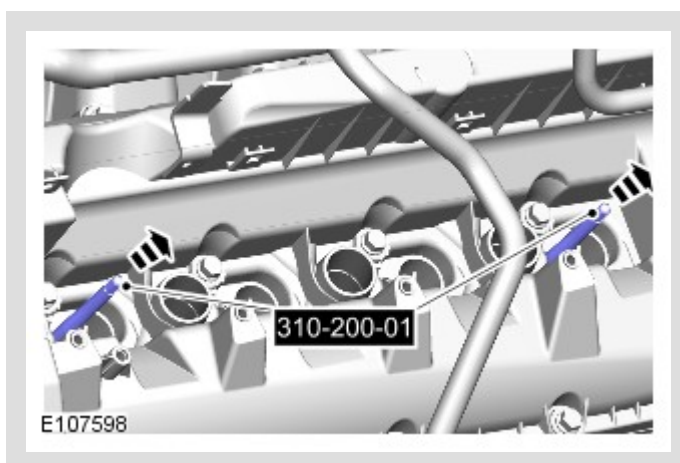


Torque: 20 Nm

16.

 **NOTE:**

LH illustration shown, RH is similar.

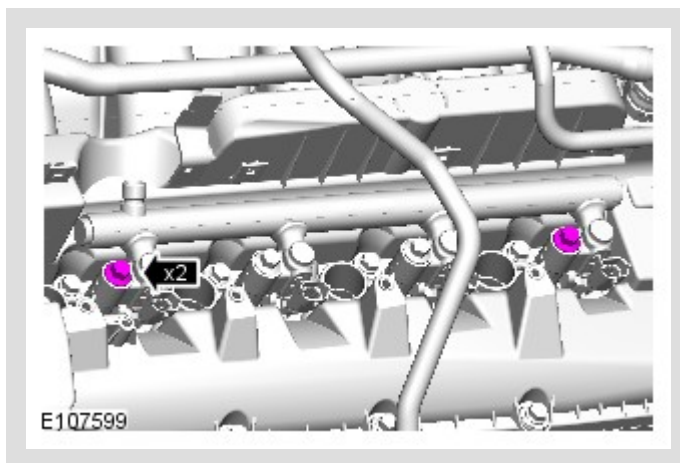


Special Tool(s): [310-200-01](#)

17.

 **NOTE:**

LH illustration shown, RH is similar.



Torque: 20 Nm

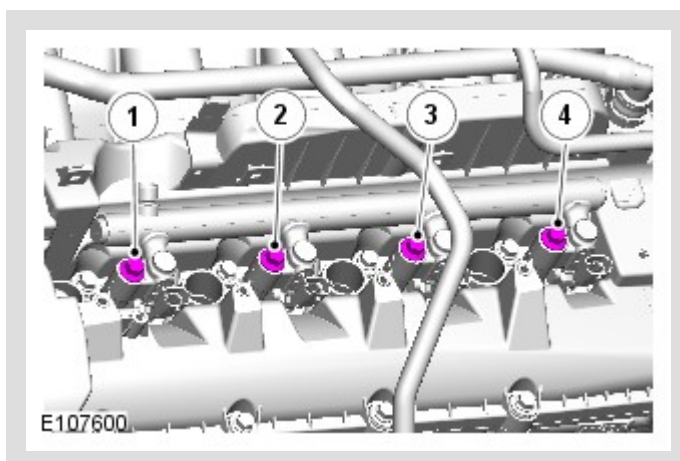
18.

⚠ CAUTION:

Tighten the bolts in the sequence shown.

△ NOTE:

LH illustration shown, RH is similar.



Special Tool(s): [310-200-01](#)

Torque:

Bolt 2 **30 Nm**

Bolt 3 **30 Nm**

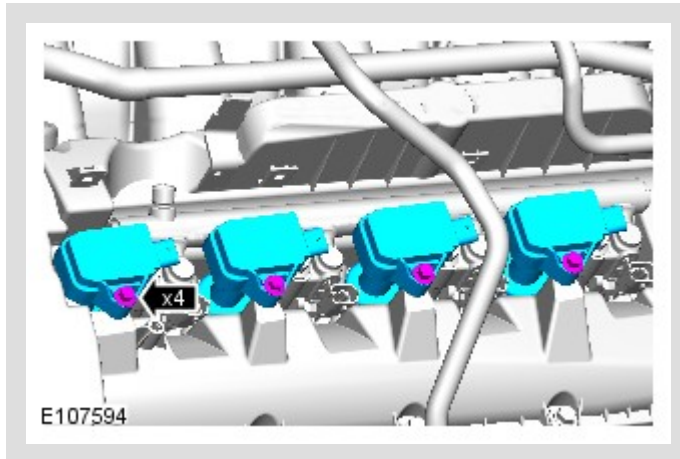
Bolt 1 **30 Nm**

Bolt 4 **30 Nm**

19.

 NOTE:

LH illustration shown, RH is similar.

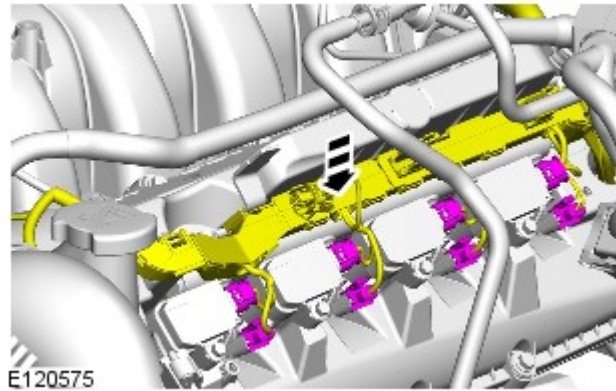


Torque: **8 Nm**

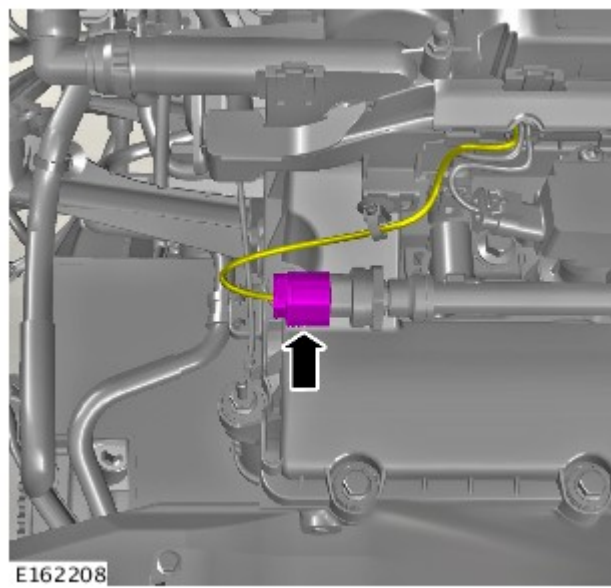
20.

 NOTE:

LH illustration shown, RH is similar.



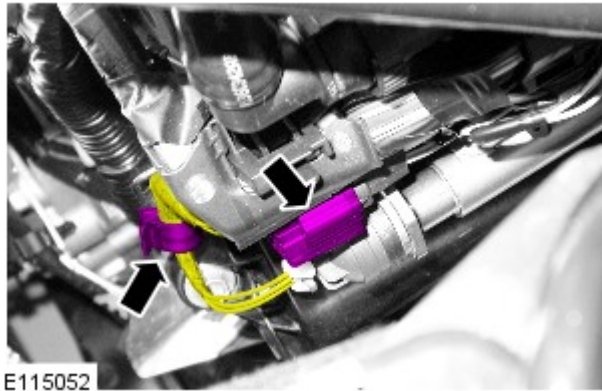
21.



22.

 **NOTE:**

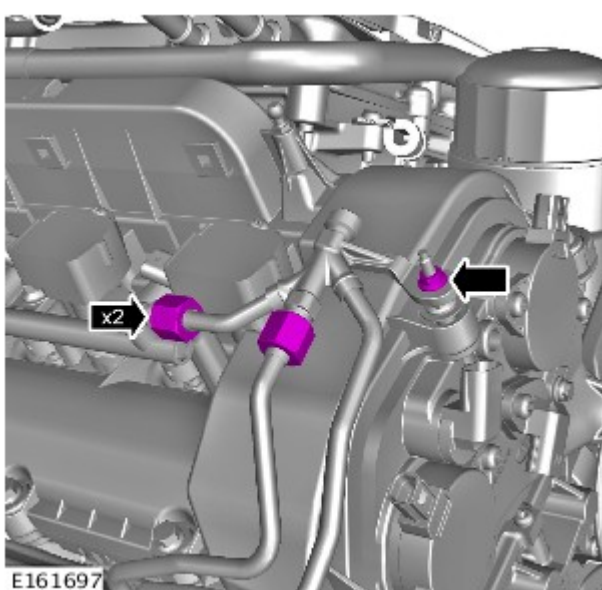
If equipped.



23.

 NOTE:

Remove and discard the blanking caps.



Torque:

Unions **21 Nm**

Nut 6 Nm

24.



WARNING:

Do not work on or under a vehicle supported only by a jack. Always support the vehicle on safety stands.

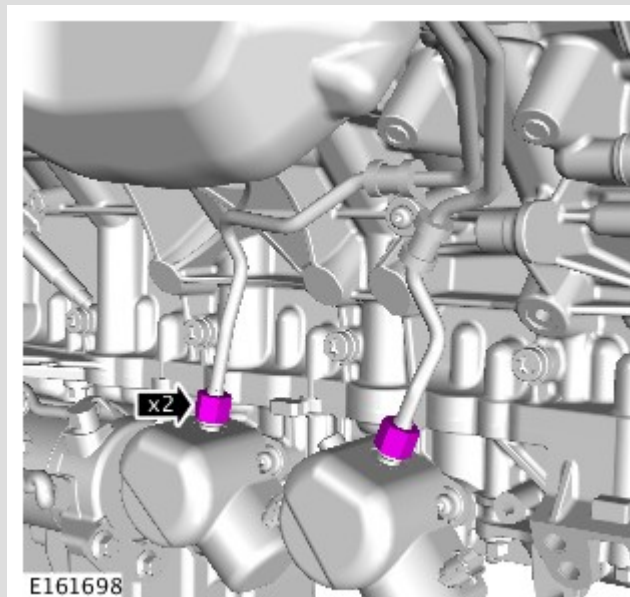
Raise and support the vehicle.

25.



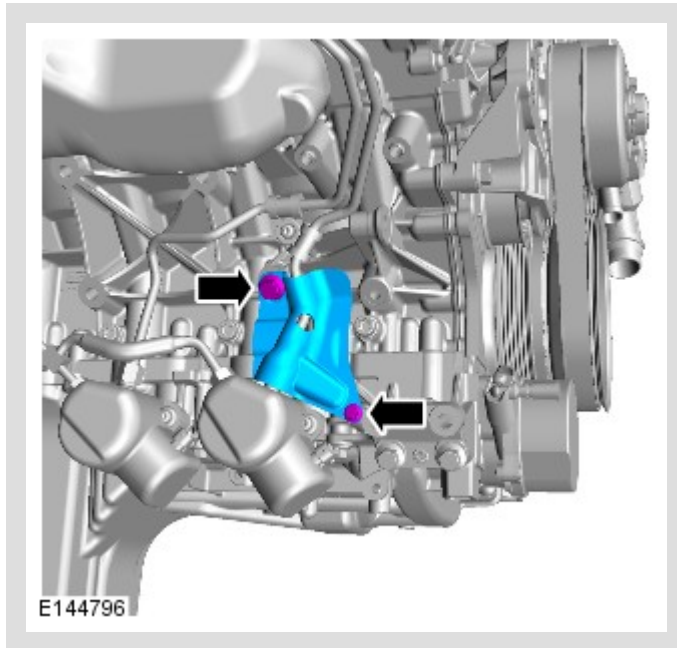
NOTE:

Some components shown removed for clarity.



Torque: 21 Nm

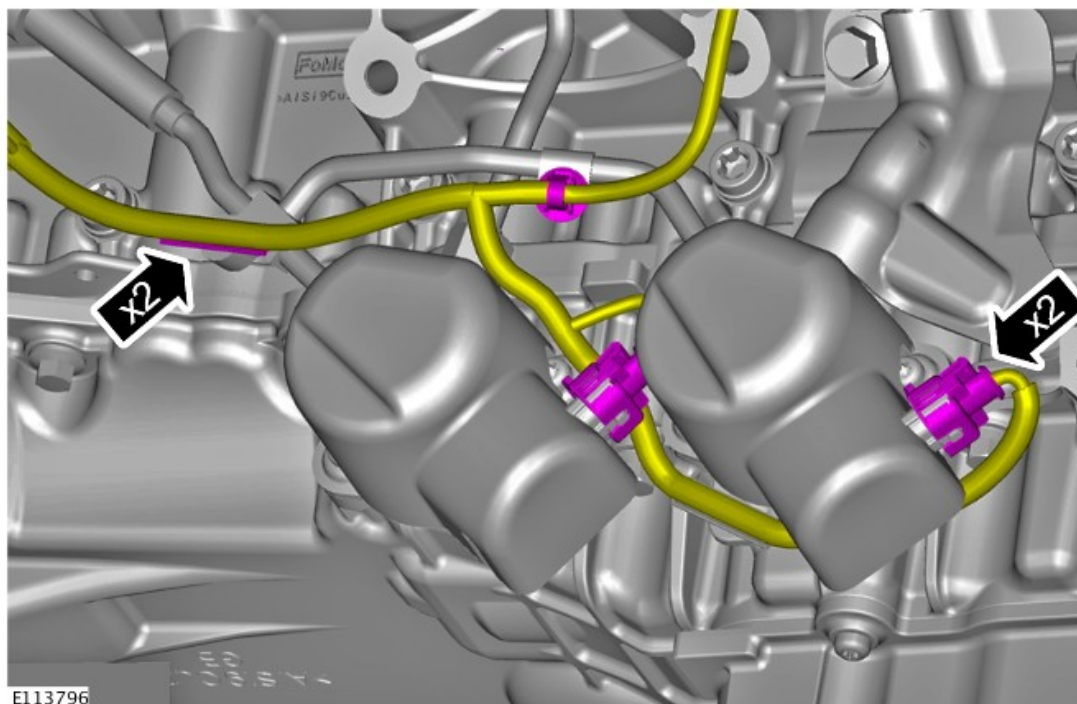
26.

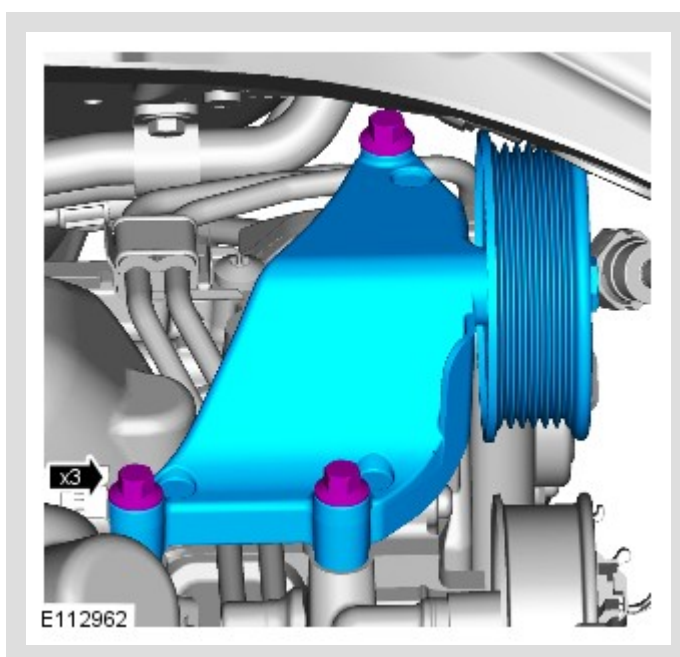


Torque:

M6 12 Nm

M10 29 Nm



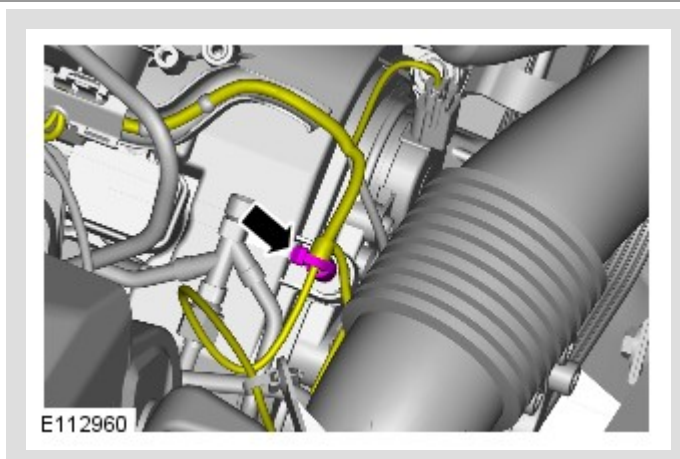


Torque: 25 Nm

29. Refer to: Generator (414-02 Generator and Regulator - V6 S/C 3.0L Petrol, Removal and Installation).

30. Lower the vehicle.

31.



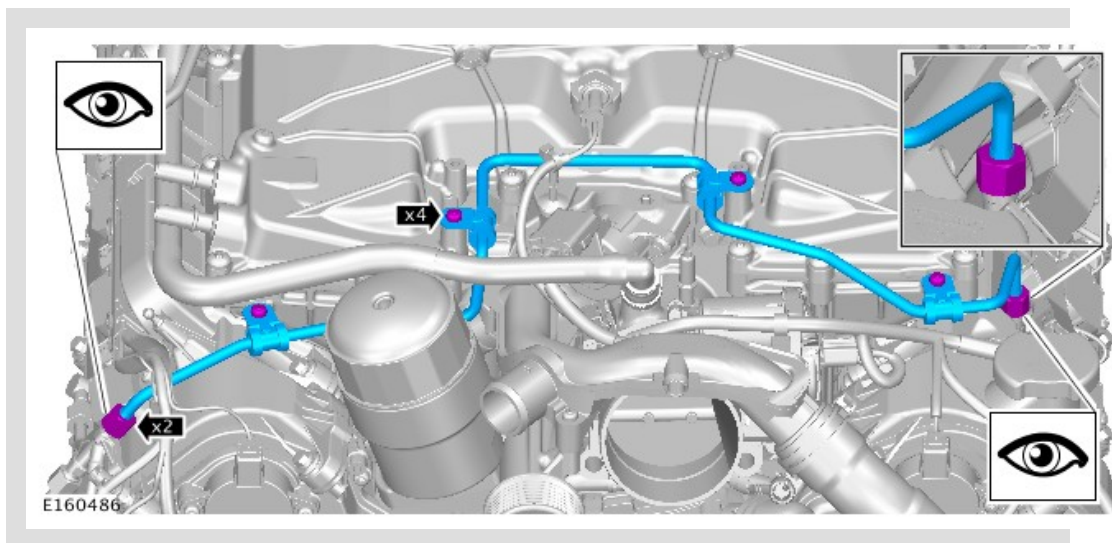
32.

**CAUTION:**

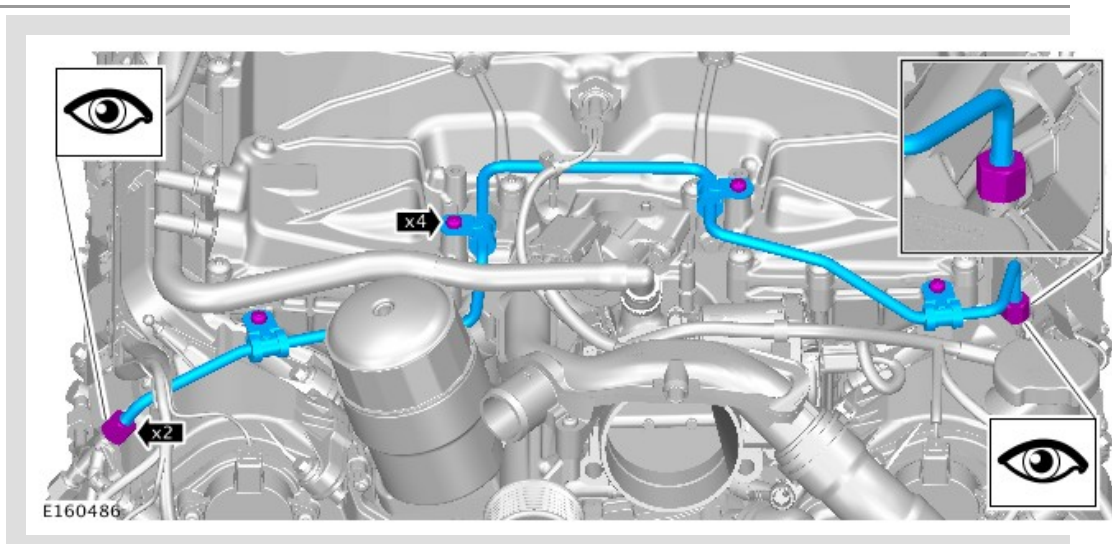
Lubricate only the union threads with clean engine oil.

**NOTES:**

- Install the bolts and unions finger tight before final tightening.
- Remove and discard the blanking caps.



33.

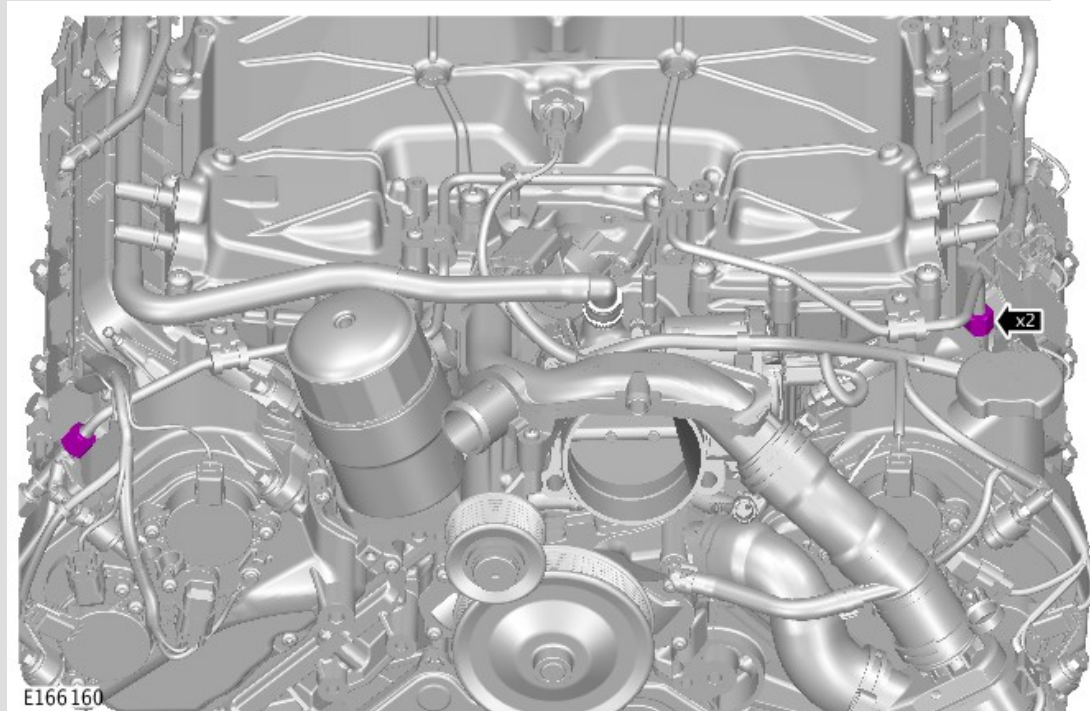


Torque:

Unions **21 Nm**

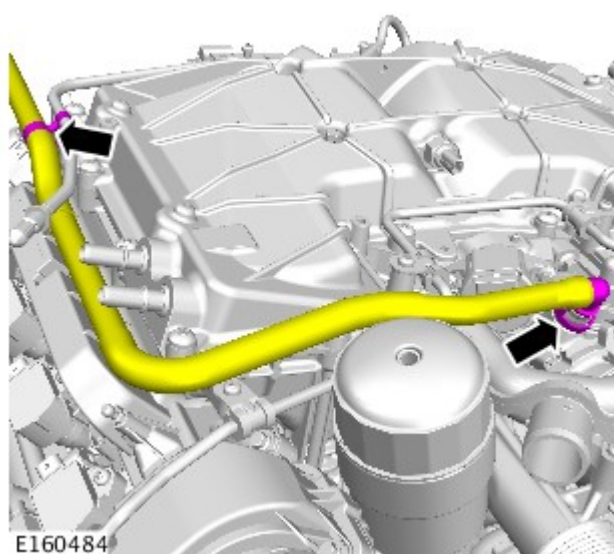
Bolts **12 Nm**

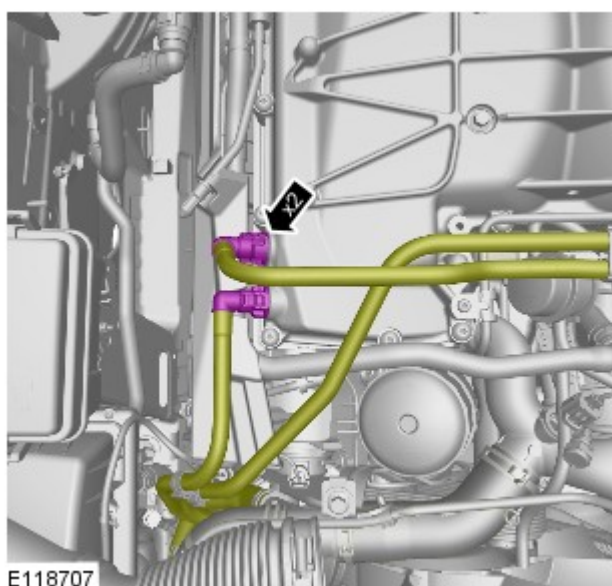
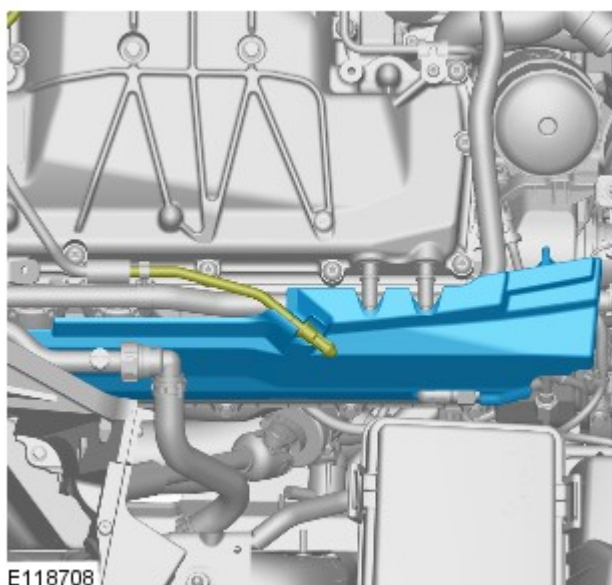
34.

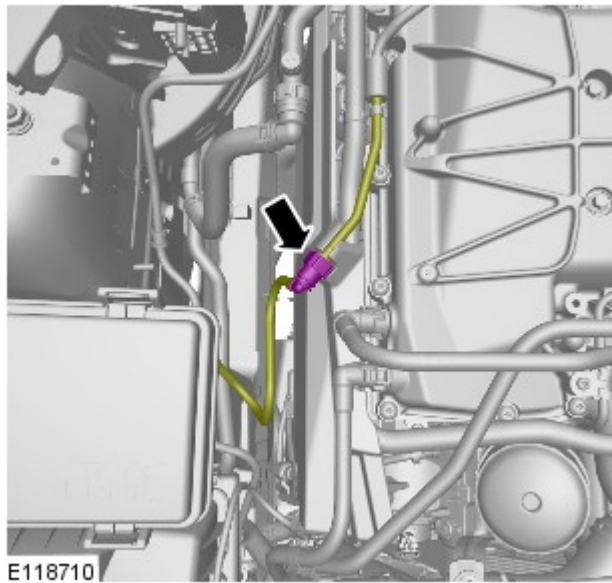


Torque: 21 Nm

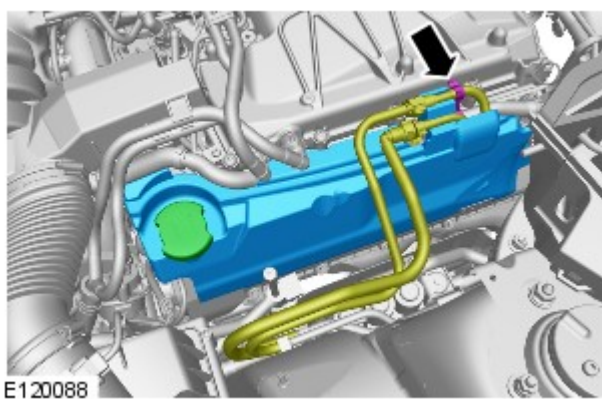
35.



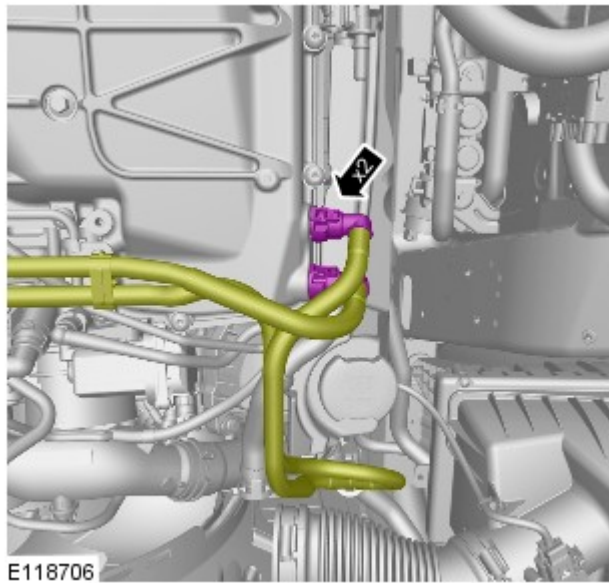




39.

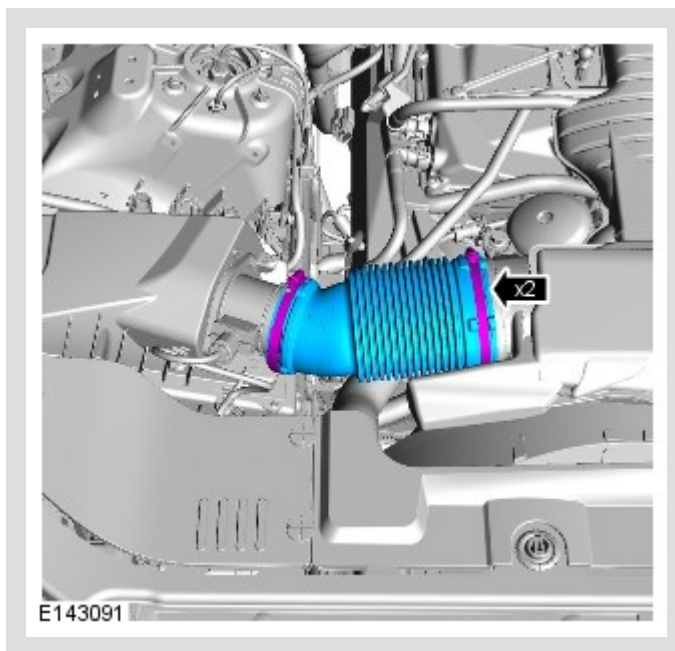


40.



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41. Refer to: Right Air Cleaner (303-12 Intake Air Distribution and Filtering - V8 S /C 5.0L Petrol, Removal and Installation).

42.



Torque: 3.5 Nm

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| 43. | Refer to: Air Cleaner Outlet Pipe T-Connector (303-12 Intake Air Distribution and Filtering - V8 S/C 5.0L Petrol, Removal and Installation). |
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| 44. | Connect the battery ground cable.
Refer to: Specifications (414-00 Battery and Charging System - General Information, Specifications). |
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| 45. | Refer to: Engine Cover - V6 S/C 3.0L Petrol /V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol (501-05 Interior Trim and Ornamentation, Removal and Installation). |
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